

A Bypass for EXCHANGE

With P2P networks, you can exchange files directly between the participating computers and avoid central servers

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Peer-to-peer (P2P) technology came to the forefront of popular consciousness when Napster's music sharing network became a rage, and drew attention for various reasons. While the future of Napster may be uncertain, the underlying peer-to-peer technology is poised to change the way we use networks.

A P2P network essentially consists of several computers (or peers) that directly exchange data among themselves. This is quite unlike a client-server model where all clients (or computers) get data from a single central server. The functions of searching for data and the actual exchange of data take place via software that is installed on the participating machines.

Napster: Search on server, share with peer

There are several different ways in which P2P technology can be implemented. In the case of Napster, the software uses your Internet connection to log you into a central Napster server.

The purpose of the server is to maintain an index of all users online at any particular moment, and help them connect to each other. It also ensures that every user on the network has shared a directory on their computer that other remote users can access, and that only MP3 files are exchanged.

When you key in a search for a particular title or artist, the client software queries the index server to find a match. When a match is found, the server displays a list of all systems that have the file in the results window. Clicking on the file establishes a direct connection between your computer and the system that stores the selected file. On successful connection, the file begins downloading. The connection is broken once the download is complete.

Gnutella: No server, just peers

In the case of Gnutella however, which is an open source P2P file sharing network, all transfers happen without a central server. Each Gnutella client combines the functions of a search engine and a file-serving agent. When you enter a search for information, the client software goes and searches for other